SUPPLEMENTAL MATERIAL

Data S1. The search method used for PubMed.

#1,"Search ""Hypertension, Pulmonary""[Mesh]",35554

#2,"Search ""Survival""[Mesh]",4690

#3,"Search ""Mortality""[Mesh]",375640

#4, "Search ((((((((((borderline pulmonary hypertension[Title/Abstract]) OR borderline pulmonary arterial pressure[Title/Abstract]) OR mild pulmonary hypertension[Title/Abstract]) OR mild elevated mean pulmonary artery pressure[Title/Abstract]) OR mild elevated mean pulmonary arterial pressure[Title/Abstract]) OR borderline pulmonary artery pressure[Title/Abstract]) OR mild elevation of pulmonary artery pressure[Title/Abstract]) OR mild elevation of pulmonary arterial pressure[Title/Abstract]) OR mild elevation of mPAP[Title/Abstract]) OR mild elevated mPAP[Title/Abstract]) OR borderline mPAP[Title/Abstract]) OR mild pulmonary arterial hypertension[Title/Abstract]",280

#5,"Search ((mortality[Title/Abstract]) OR Mortalities[Title/Abstract]) OR Case Fatality Rate[Title/Abstract]) OR Case Fatality Rates[Title/Abstract]) OR Rate, Case Fatality[Title/Abstract]) OR Rates, Case Fatality[Title/Abstract]) OR Mortality, Excess[Title/Abstract]) OR Excess Mortalities[Title/Abstract]) OR Mortalities, Excess[Title/Abstract]) OR Excess Mortality[Title/Abstract]) OR Decline, Mortality[Title/Abstract]) OR Declines, Mortality[Title/Abstract]) OR Mortality Declines[Title/Abstract]) OR Mortality Decline[Title/Abstract]) OR Mortality Determinants[Title/Abstract]) OR Determinant, Mortality[Title/Abstract]) OR Mortality Determinant[Title/Abstract]) OR Determinants,

Mortality[Title/Abstract]) OR Mortality, Differential[Title/Abstract]) OR Differential Mortalities[Title/Abstract]) OR Mortalities, Differential[Title/Abstract]) OR Differential Mortality[Title/Abstract]) OR Age-Specific Death Rate[Title/Abstract]) OR Age-Specific Death Rates[Title/Abstract]) OR Death Rate, Age-Specific[Title/Abstract]) OR Death Rates, Age-Specific[Title/Abstract]) OR Rate, Age-Specific Death[Title/Abstract]) OR Rates, Age-Specific Death[Title/Abstract]) OR Age Specific Death Rate[Title/Abstract]) OR Death Rate[Title/Abstract]) OR Death Rates[Title/Abstract]) OR Rate, Death[Title/Abstract]) OR Rates, Death[Title/Abstract]) OR Mortality Rate[Title/Abstract]) OR Mortality Rates[Title/Abstract]) OR Rate, Mortality[Title/Abstract]) OR Rates, Mortality[Title/Abstract]) OR survival[Title/Abstract]) OR pulmonary hypertension[Title/Abstract]) OR hazard ratio[Title/Abstract]) OR ""Mortality""[Mesh]) OR ""Survival""[Mesh]) OR ""Hypertension, Pulmonary""[Mesh]",1823256

#6,"Search ((((((((((((borderline pulmonary hypertension[Title/Abstract]) OR borderline pulmonary arterial pressure[Title/Abstract]) OR mild pulmonary hypertension[Title/Abstract]) OR mild elevated mean pulmonary artery pressure[Title/Abstract]) OR mild elevated mean pulmonary arterial pressure[Title/Abstract]) OR borderline pulmonary artery pressure[Title/Abstract]) OR mild elevation of pulmonary artery pressure[Title/Abstract]) OR mild elevation of pulmonary arterial pressure[Title/Abstract]) OR mild elevation of mPAP[Title/Abstract]) OR mild elevated mPAP[Title/Abstract]) OR borderline mPAP[Title/Abstract]) OR mild pulmonary arterial hypertension[Title/Abstract])) AND ((mortality[Title/Abstract]) OR Mortalities[Title/Abstract]) OR Case Fatality Rate[Title/Abstract]) OR Case Fatality Rates[Title/Abstract]) OR Rate, Case

Fatality[Title/Abstract]) OR Rates, Case Fatality[Title/Abstract]) OR Mortality, Excess[Title/Abstract]) OR Excess Mortalities[Title/Abstract]) OR Mortalities, Excess[Title/Abstract]) OR Excess Mortality[Title/Abstract]) OR Decline, Mortality[Title/Abstract]) OR Declines, Mortality[Title/Abstract]) OR Mortality Declines[Title/Abstract]) OR Mortality Decline[Title/Abstract]) OR Mortality Determinants[Title/Abstract]) OR Determinant, Mortality[Title/Abstract]) OR Mortality Determinant[Title/Abstract]) OR Determinants, Mortality[Title/Abstract]) OR Mortality, Differential[Title/Abstract]) OR Differential Mortalities[Title/Abstract]) OR Mortalities, Differential[Title/Abstract]) OR Differential Mortality[Title/Abstract]) OR Age-Specific Death Rate[Title/Abstract]) OR Age-Specific Death Rates[Title/Abstract]) OR Death Rate, Age-Specific[Title/Abstract]) OR Death Rates, Age-Specific[Title/Abstract]) OR Rate, Age-Specific Death[Title/Abstract]) OR Rates, Age-Specific Death[Title/Abstract]) OR Age Specific Death Rate[Title/Abstract]) OR Death Rate[Title/Abstract]) OR Death Rates[Title/Abstract]) OR Rate, Death[Title/Abstract]) OR Rates, Death[Title/Abstract]) OR Mortality Rate[Title/Abstract]) OR Mortality Rates[Title/Abstract]) OR Rate, Mortality[Title/Abstract]) OR Rates, Mortality[Title/Abstract]) OR survival[Title/Abstract]) OR pulmonary hypertension[Title/Abstract]) OR hazard ratio[Title/Abstract]) OR ""Mortality""[Mesh]) OR ""Survival""[Mesh]) OR ""Hypertension, Pulmonary""[Mesh])",248

Table S1. Quality assessment of the cohort studies included in the meta-analysis using the Newcastle Ottawa Scale (NOS). A study can be awarded a maximum of one star for each numbered item within the Selection and Outcome categories. A maximum of two stars can be given for Comparability

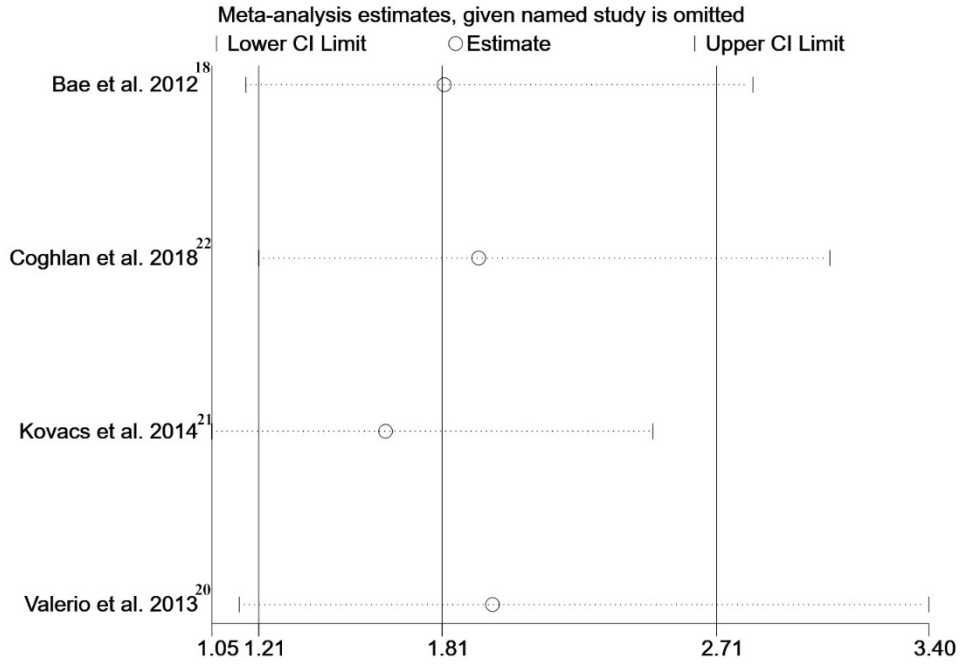
Study		Selection					Outcome			
Author	Year	Representativeness of the exposed cohort	Selection of the non-exposed cohort	Ascertainment of exposure	outcome of interest was not present at start of study	Comparability	Assessment of outcome	Adequate follow-up period for outcome	Adequacy of follow up of cohorts	Total Score
Bae et al.[18]	2012	*	*	*	*	*	*	*	*	8
Heresi et al.[19]	2013	*	*	*	*	**	*	*	*	9
Valerio et al.[20]	2013	*	*	*	*	*	*	*	*	8
Kovacs et al.[21]	2014	*	*	*	*		*	*	*	7
Coghlan et al.[22]	2018	*	*	*	*		*	*	*	7
Douschan et al.[23]	2018	*	*	*	*		*	*	*	7
Nemoto et al.[24]	2019	*	*	*	*	**	*		*	8
Xanthouli et al.[25]	2019	*	*	*	*	**	*	*	*	9

Table S2. The pooled survival rates and their 95% CIs of the normal PAP, mildly elevated PAP and PH groups.

Year	The Pooled Survival Rates (95%CI), %		
	Normal PAP	Mildly elevated PAP	PH
1	97.7 (95.4-100.0)	97.0 (93.7-100.0)	87.5 (80.9-94.7)
2	95.9 (93.6-98.4)	92.8 (88.6-97.2)	80.2 (70.8-90.8)
3	93.3 (90.2-96.5)	89.4 (84.0-95.2)	73.0 (63.6-83.9)
4	91.0 (86.2-96.0)	80.7 (72.3-90.1)	65.7 (56.2-76.9)
5	88.8 (83.4-94.6)	77.0 (67.2-88.3)	61.6 (51.9-73.1)
6	85.2 (77.5-93.5)	70.6 (62.4-79.8)	54.8 (47.9-62.6)
7	81.0 (72.3-90.7)	64.5 (55.4-75.0)	50.0 (43.9-57.1)
8	75.5 (68.5-83.3)	58.8 (46.9-73.6)	46.7 (40.4-54.0)
9	74.9 (67.8-82.8)	49.6 (35.5-69.4)	41.3 (34.8-48.9)

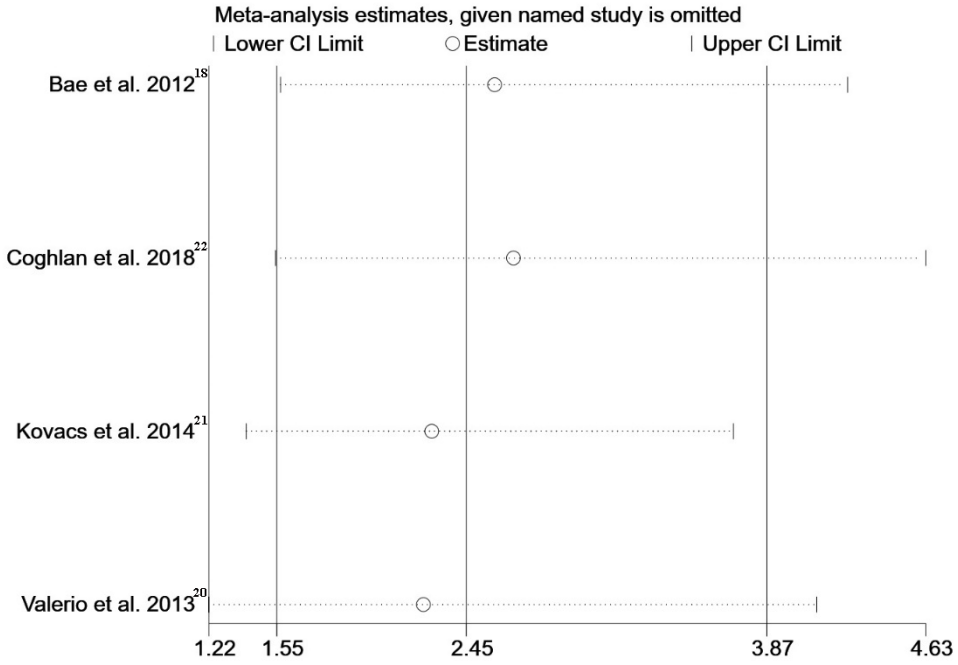
PAP: pulmonary artery pressure; PH: pulmonary hypertension; CI: confidence interval.

Figure S1. Sensitivity analysis for the meta-analysis of risk of progression to PH in the mildly elevated PAP group versus the normal PAP group when only participants who had repeated RHCs were included in the calculation of the incidence of progression to PH (the analysis shown in Figure 2A).



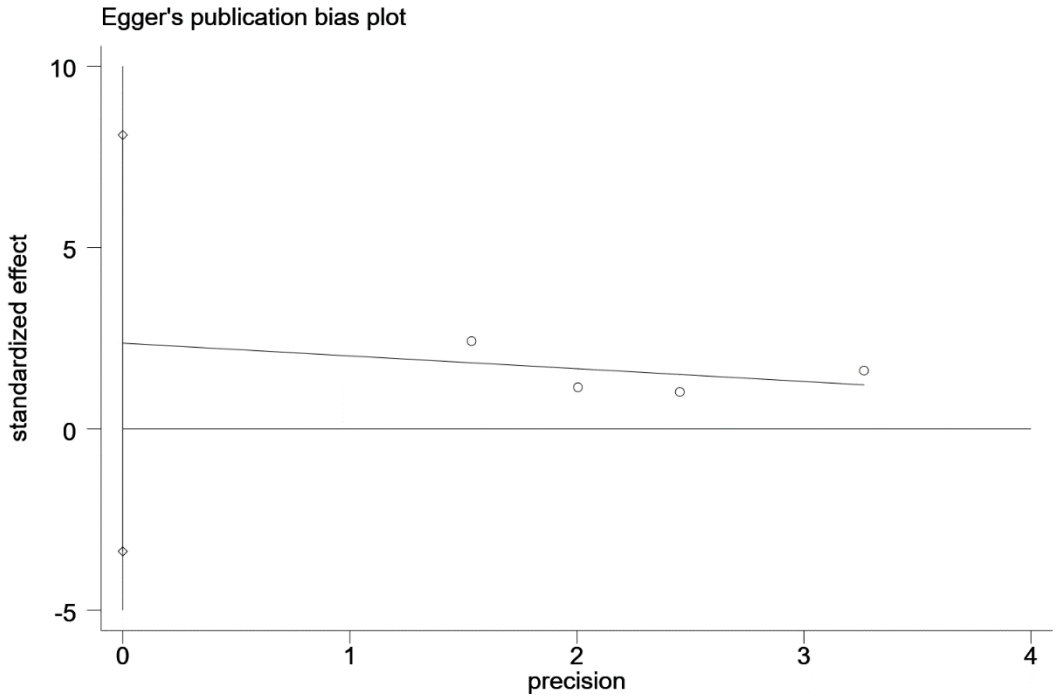
PH: pulmonary hypertension; PAP: pulmonary artery pressure; RHC: right heart catheterization.

Figure S2. Sensitivity analysis for the meta-analysis of risk of progression to PH in the mildly elevated PAP group versus the normal PAP group when all participants in each group were included in the total number (the analysis shown in Figure 2B).



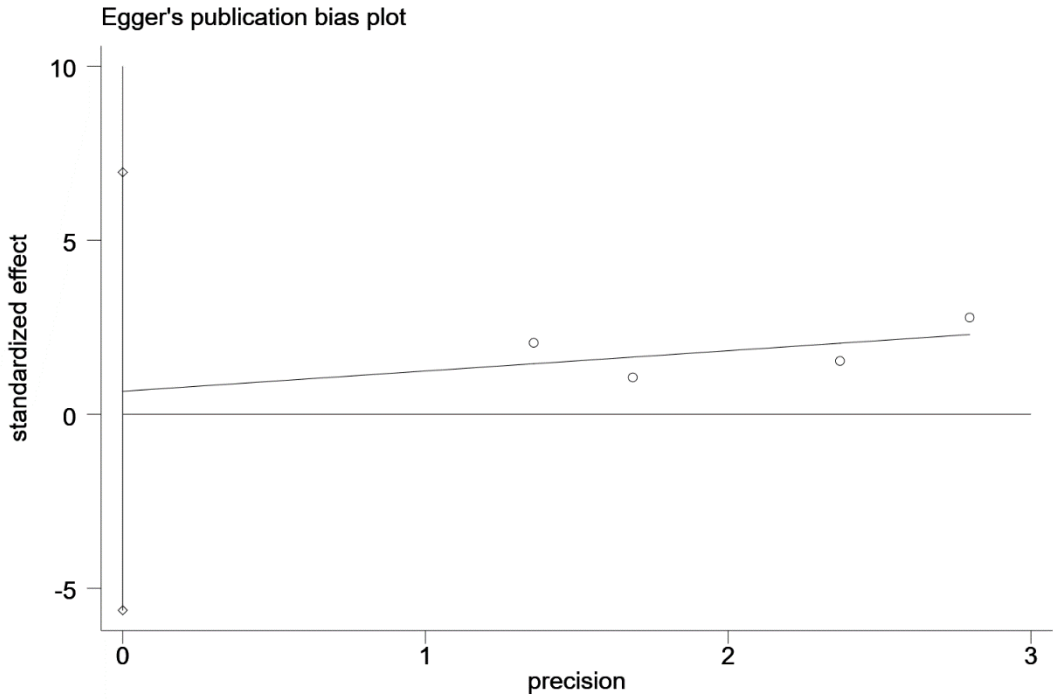
PH: pulmonary hypertension; PAP: pulmonary artery pressure.

Figure S3. Egger's test for the meta-analysis of risk of progression to PH in the mildly elevated PAP group versus the normal PAP group when only participants who had repeated RHCs were included in the calculation of the incidence of progression to PH (the analysis shown in Figure 2A), *P*=0.218.



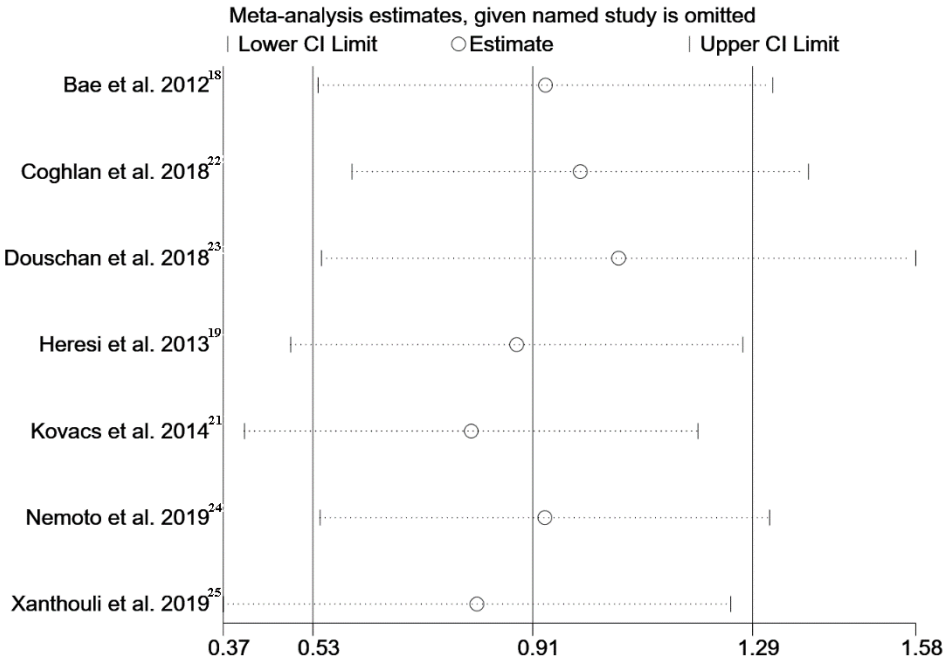
PH: pulmonary hypertension; PAP: pulmonary artery pressure; RHC: right heart catheterization.

Figure S4. Egger's test for the meta-analysis of risk of progression to PH in the mildly elevated PAP group versus the normal PAP group when all participants in each group were included in the total number (the analysis shown in Figure 2B), *P*=0.695.



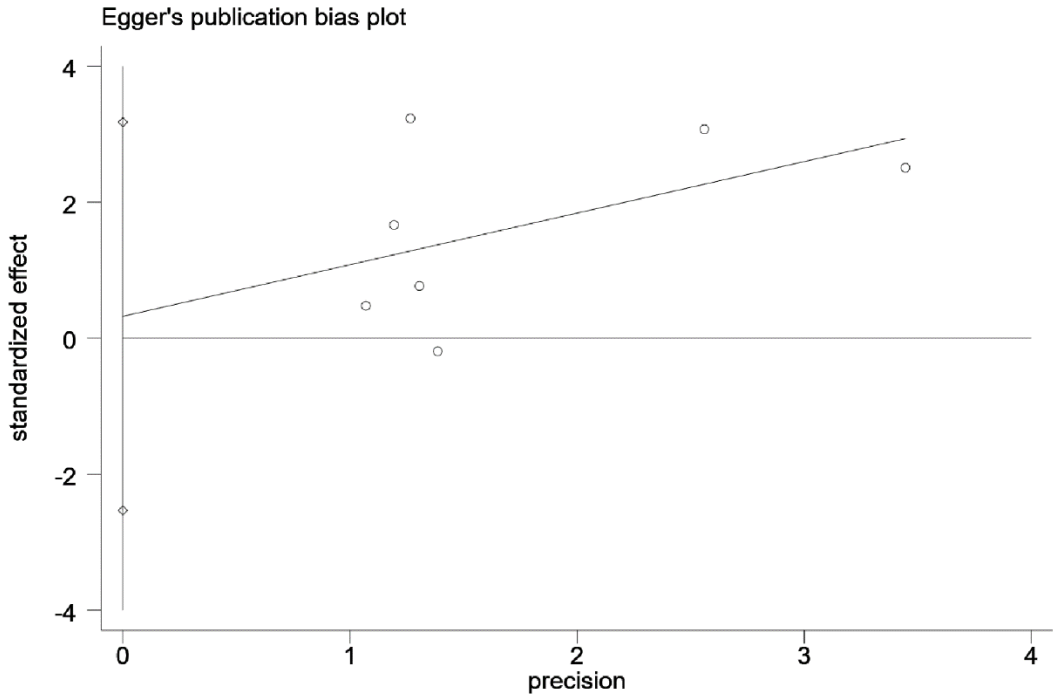
PH: pulmonary hypertension; PAP: pulmonary artery pressure.

Figure S5. Sensitivity analysis for the meta-analysis of pooled hazard ratio for mortality in the mildly elevated PAP group versus the normal PAP group (the analysis shown in Figure 3).



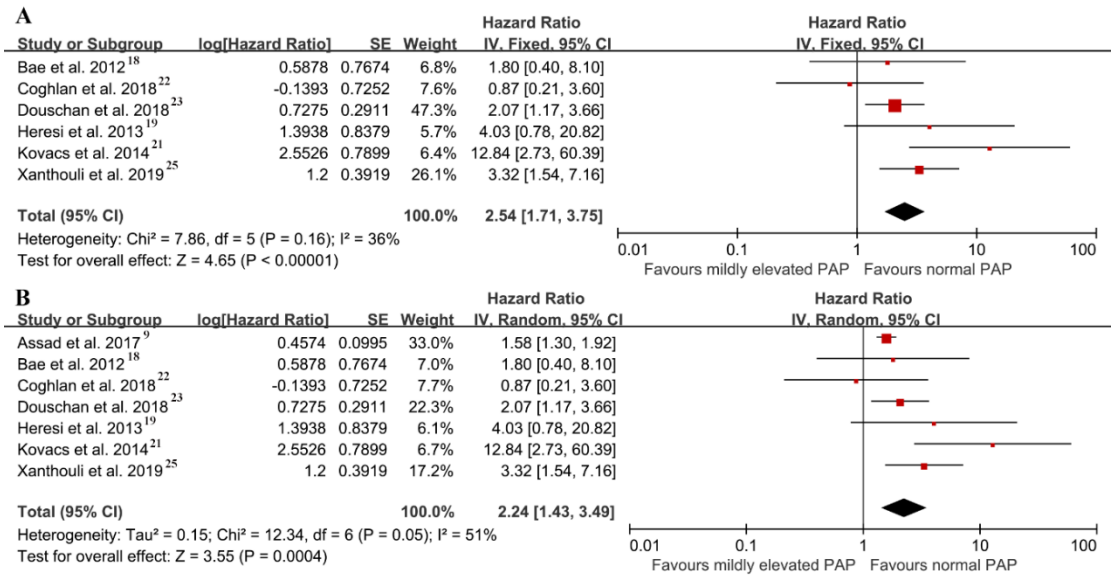
PAP: pulmonary artery pressure.

Figure S6. Egger's test for the meta-analysis of pooled hazard ratio for mortality in the mildly elevated PAP group versus the normal PAP group (the analysis shown in Figure 3), *P*=0.861.



PAP: pulmonary artery pressure.

Figure S7. Supplementary analyses of pooled hazard ratio for mortality in the mildly elevated PAP group versus the normal PAP group.



A: Supplementary analysis when Nemoto's study was excluded (fixed-effects model); B: Supplementary analysis when Assad's study was added while Nemoto's study was excluded (random-effects model). PAP: pulmonary artery pressure; IV: inverse variance; CI: confidence interval.